

Isaac Gallegos

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RESEARCH INTERESTS

quantum information · quantum networks, photonic interconnects, and entanglement distribution between spin-based quantum nodes · many-body quantum systems and statistical physics.

EDUCATION

Colorado School of Mines Golden, CO
B.S., *Engineering Physics*; GPA: 3.99/4.00, *summa cum laude* Dec 2024

- Senior Thesis: *Emergent Holography from Information Geometry of Ising Models*. Advisor: L. Haddad
- Concentration: Quantum Science, Laser Physics and Engineering, Electrical Engineering
- Graduate-level and advanced coursework: Graduate Laser Physics; Quantum Mechanics I & II; Electromagnetic Waves and Optical Physics; Digital and Analog Electronics; Mathematical Methods of Physics

HONORS & AWARDS

- Dean's List, Colorado School of Mines (2021–2024)
- Presidential Scholarship, Colorado School of Mines (2021–2024)
- Coors Foundation Scholarship (2021–2024)
- Fellowship applications in submission for 2027 admissions cycle: NSF Graduate Research Fellowship Program (GRFP), Hertz Foundation Fellowship, Department of Defense NDSEG Fellowship

RESEARCH EXPERIENCE

Harvard Medical School – Wellman Center for Photomedicine (Bouma Lab) Boston, MA
Research Engineer May 2025–Present

Advisors: D. Veyssset, N. Uribe-Patarroyo, B.E. Bouma

- Leading a polarization-resolved spectroscopic OCT project using a 1310 nm swept-source VCSEL system (100 nm bandwidth, 5 μ m axial resolution). Acquired and spectrally post-processed data with 20+ fabricated biomaterial phantoms
- Wrote physics-constrained inverse models in Python to recover wavelength-dependent optical properties from depth-resolved spectra
- Built the first comprehensive open-source library and software tool for laser safety evaluation; first author on a manuscript targeting the *Journal of Open Source Software*

Research Assistant Feb 2025–May 2025

Advisors: D. Veyssset, B.E. Bouma

- Built the first micro-LED-based OCT system prototype, replacing swept-source lasers for individually addressable narrowband LEDs, and wrote its hardware control and acquisition software
- Applied spectral unmixing to multi-spectral photoacoustic data from a custom, all-fiber stimulated Raman scattering optical amplifier, determining per-pixel lipid and collagen concentrations from imaged samples

Colorado School of Mines – Physics Department Golden, CO
Undergraduate Research Assistant Aug 2023–May 2024

Advisor: L. Haddad

- Developed an information-geometry approach to emergent holography, reparameterizing the 1D Ising model through a Gaussian dressing whose Fisher metric yields an emergent AdS_2 metric

- Evaluated the partition function by transfer-matrix methods and worked out the dressing's two limiting regimes as dual descriptions of the lattice; first author on a manuscript in preparation for *JHEP*

Colorado School of Mines – Ultrafast Laser Research Laboratory

Golden, CO

Undergraduate Research Project Lead

Aug 2023–Dec 2023

Advisor: C. Durfee

- Led a five-person undergraduate team designing micron-scale, supersonic gas-jet nozzles for high-harmonic generation experiments, designing 5+ converging-diverging geometries in SolidWorks and Mathematica resulting in an optimized configuration

SELECTED PROJECTS**Polarization-Entangled Photon Pair Generation & Bell-Inequality Measurement**

Golden, CO

Project Member, Graduate Quantum Optics Laboratory

Aug 2024–Dec 2024

- Produced polarization-entangled 810 nm photon pairs from dual-crystal Type-I SPDC under a 405 nm diode pump, using a YVO₄ crystal to compensate birefringent walk-off
- Measured a CHSH violation of $S = 2.51 \pm 0.04$, confirming non-local entanglement, with silicon SPADs and four-channel FPGA coincidence counting with nanosecond gating

Solid-State Laser Construction and Rate-Equation Modeling

Golden, CO

Project Lead, Graduate Laser Physics Laboratory

Aug 2024–Dec 2024

- Designed, built, and characterized a CW diode-pumped Pr³⁺:LiYF₄ laser (640 nm, 35 mW, 18% slope efficiency)
- Checked four-level rate-equation models against the measured pulse trains, fluorescence spectra, and output power

Diode-Pumped Q-Switched Nd:YAG Laser

Golden, CO

Project Lead, Graduate Laser Physics Laboratory

Aug 2024–Dec 2024

- Demonstrated short-pulse generation by passively Q-switching an Nd:YAG laser (1064 nm, Cr:YAG saturable absorber, repetition rates to 3.5 kHz)
- Computationally modeled Q-switching in Mathematica to verify experimental results; characterized pulse repetition rates, durations, and energy to demonstrate laser functionality

Software & Open-Source Tools*Personally developed and publicly released*

2024–Present

- **Maximum Permissible Exposure tool:** open-source software that computes safety limits for pulsed and CW laser sources, with a graphical interface for laboratory use
- **isaac-gallegos.com:** production Astro 6 + MDX + Tailwind 4 academic site with BibTeX citations and Pagefind search

PUBLICATIONS & PRESENTATIONS*Manuscripts in Preparation*

1. **Gallegos, I.**, McAuley, R., Borges, N., Bouma, B.E., Veyssset, D. "Maximum Permissible Exposure Tool for Laser Safety Evaluation." Target: *JOSS*, 2026
2. **Gallegos, I.**, Carmody, E., Nguyen, D., Haddad, L. "Emergent holography from information geometry of Ising models." Target: *Journal of High Energy Physics*, 2026

Conference Presentations

1. **Gallegos, I.**, Veyssset, D., Bouma, B.E. "Multi-spectral NIR-II photoacoustic microscopy utilizing a custom all-fiber amplifier." Poster, *Celebration of Science*, Massachusetts General Hospital, 2025.
2. **Gallegos, I.**, Ruiz-Lopera, S., McAuley, R., Bouma, B.E., Veyssset, D., Uribe-Patarroyo, N. "Quantitative spectroscopic OCT for compositional characterization of coronary plaques." Poster, *NESBO*, MIT, 2025.

3. De León Barba, A., Norman, N., Sahasrabudhe, A., Aboukhalil, K., **Gallegos, I.**, Bouma, B.E., Kim, J., Uribe-Patarroyo, N., Veysset, D. "Spectroscopic OCT with multichromatic micro-LED illumination." Poster, *NESBO*, MIT, 2025.
4. Aboukhalil, K., Sahasrabudhe, A., **Gallegos, I.**, Bouma, B.E., Kim, J., Uribe-Patarroyo, N., Veysset, D. "Proof-of-concept visible-light OCT using micro-LED arrays." Poster, *SPIE OPTO*, San Francisco, 2026.

TECHNICAL SKILLS

Quantum Optics	SPDC photon-pair generation; single-photon detection (silicon SPADs, time-correlated coincidence counting); polarization-resolved interferometry and Jones-matrix characterization; Bell-state measurement; broadband and Raman spectroscopy
Computation	Python (PyTorch, JAX, NumPy, SciPy); MATLAB; C++; Mathematica. Differentiable physics modeling, physics-constrained inverse problems
Photonic Simulation	COMSOL Multiphysics; Zemax OpticStudio; SolidWorks; MEEP
Lasers & Optics	Solid-state laser construction ($\text{Pr}^{3+}:\text{LiYF}_4$); free-space and fiber optics; optomechanical assembly; optical coherence tomography (swept-source, micro-LED illumination, polarization-sensitive)
Electronics	FPGA design and gateway (Altera Quartus Prime); time-correlated single-photon counting; PCB design and fabrication; analog and digital circuit design
Cleanroom & Materials	Clean-room fabrication, thin-film deposition (sputtering, thermal evaporation), vacuum systems, ellipsometry, SEM

TEACHING

Colorado School of Mines – Physics Department

Golden, CO

Teaching Assistant

Aug 2023–Dec 2024

Courses: *Electromagnetic Waves and Optical Physics* (two semesters) · *Mathematical Physics* · *Intermediate Electromagnetism*

- Provided supplemental lectures, review sessions, and office hours across four semesters of upper-division physics, writing original exams and problem sets and mentoring 50+ undergraduates

MENTORING & LEADERSHIP

Reign Tutoring

Golden, CO

Founder & Tutor

Aug 2021–Aug 2023

- Founded and ran a private tutoring company for two years, working with 15+ students in physics, mathematics, and engineering

PROFESSIONAL AFFILIATIONS

SPIE · Optica · IEEE Photonics Society · Society of Hispanic Professional Engineers (SHPE).

LANGUAGES & INTERESTS

English (Native) · Mandarin Chinese (elementary) · speleology & vertical caving · trail running

REFERENCES

Brett E. Bouma: Professor, MIT; Wellman Center for Photomedicine, HMS, bouma@mgh.harvard.edu

Nestor Uribe-Patarroyo: Assistant Professor, Wellman Center for Photomedicine, HMS, UribePatarroyo.Nestor@mgh.harvard.edu

David Veysset: Instructor, Wellman Center for Photomedicine, HMS, dveysset@mgh.harvard.edu

Laith Haddad: Teaching Assistant Professor, Colorado School of Mines, lhaddad@mines.edu

Jeff Squier: Professor, Colorado School of Mines, jsquier@mines.edu